Air Convection

Student sheet

Name: _____ **Date:** _____

Safety: The teacher pours the warm water. Do not touch the bulb or radiator. Keep the box dry.

1. What I see on the screen

The screen shows **Temperature**, **Humidity** and their chart. **Watch the shape of the curve, not just the number.** Move on with the green button and go back with **Back**. **The sensor is slow:** 60–120 s for T, up to 3 min for humidity. It stays 10–15 cm above the source, never touching it.

2. What I do on my own

Dry source (circle one): radiator / lamp / laptop vent **Heating on?** ☐ YES ☐ NO

First I predict, on the sheet and in the app. I do not change it after I measure.

1. Near the ceiling the air is ☐ warmer ☐ cooler ☐ the same as near the floor, by _____ °C.

2. The humidity will ☐ rise ☐ fall ☐ stay above the **warm water**, and will ☐ rise ☐ fall ☐ stay above the **radiator** (or the lamp),

because _____

_____

3. I draw the arrows. On the left, *before* any measurement: how do you think the air moves around the source? On the right, at the end, what you saw. **The loop must close.**

What I predict (before) **What I saw (after)**

Now I measure. Shape of the curve: rises fast / rises slowly / has stopped / falls.

Position (15 cm)	T (°C)	**RH (%)**	**Shape of the curve**
Reference, on the desk			flat
Above the warm water			
Above the dry source			
Back on the desk			

Difference (above the source − reference) = _____ − _____ = _____ °C

How sure are we? Boxes side by side: ± ____ °C. Same place, twice: ____ and ____ °C

The vertical profile. At least 90 s at every height; heights are *measured*, not guessed.

Height	T **(°C)**	**Compared to the desk**
0 cm (floor)		☐ warmer ☐ cooler
50 cm		☐ warmer ☐ cooler
100 cm		☐ warmer ☐ cooler
150 cm		☐ warmer ☐ cooler
200 cm		☐ warmer ☐ cooler

The graph. The temperature axis does not start at zero.

top: _____°C

bottom: _____°C

Height above the floor, from 0 to 200 cm — ten divisions of 20 cm each

The gradient. Top _____ − floor _____ = _____ °C, over ____ m, so _____ °C/m; in a sports hall 4 m high it would be _____ °C, but my prediction could be wrong because _____

3. The conclusion I write

I predicted that _____

_____

The data show that _____

_____

Heated air rises because _____

_____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.

_____